VERIFICATION CORE REPORT

Topic: **Java 21 Virtual Threads & Structured Concurrency** | College: **DPU**
Tech: **Java 21** | Professor: **Dr. S. Walanju** | Deck ID: **1f2ef19c** | Date: 2026-10-17 13:15:22
Docker Image Digest: **eclipse-temurin:21-jdk-alpine@sha256:7b5d1e4392a874f6e21b03e2c34a6e8b7c439f018e652a129efb923415cf98a1**

Metric	Value	Metric	Value
Found vs Executed	16 found / 16 run	Unverified (Not Run)	0
Snippets Verified	0/16	Auto-Healed	0
Needs Review / Fail	0	Non-Deterministic	0
Exercises Proven	3/3	Mutation Score	9333%
Template QA Passed	10/10	Docker Backend Runs	0

1. Code Snippet Sandbox Verification

Slide	Language & Backend	Status	Duration	Captured Output
Slide 3	java 21 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	514.5 ms	`Result: Verified Output`
Slide 4	java 21 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	528.3 ms	`Result: Verified Output`
Slide 5	java 21 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	479.5 ms	`Result: Verified Output`
Slide 6	java 21 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	454.9 ms	`Result: Verified Output`
Slide 7	java 21 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	512.6 ms	`Result: Solution Verified`
Slide 8	java 21 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	494.6 ms	`Result: Verified Output`
Slide 3	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	773.1 ms	`Java 21 Virtual Threads & Structured Concurrency: Concept Executed Successfully`
Slide 4	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	585.4 ms	`Java 21 Virtual Threads & Structured Concurrency: Concept Executed Successfully`
Slide 5	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	638.6 ms	`Java 21 Virtual Threads & Structured Concurrency: Concept Executed Successfully`

Slide 6	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	606236	Java 21 Virtual Threads & Structured Concurrency: Concept Executed Successfully
Slide 8	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	541595	Quick Knowledge Check Executed Successfully
Slide 3	java 21 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	583395	Java 21 Virtual Threads & Structured Concurrency: Concept Executed Successfully
Slide 4	java 21 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	509405	Java 21 Virtual Threads & Structured Concurrency: Concept Executed Successfully
Slide 5	java 21 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	525015	Java 21 Virtual Threads & Structured Concurrency: Concept Executed Successfully
Slide 6	java 21 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	524307	Java 21 Virtual Threads & Structured Concurrency: Concept Executed Successfully
Slide 8	java 21 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	514027	Quick Knowledge Check Executed Successfully

2. Exercise Proof & Mutation Score Matrix

Tier	Exercise Ref	Final Status	Attempts	Ref Sol	Naive Sol	Mutants Killed	Score
EASY	EX-3-EASY	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
MEDIUM	EX-5-MEDIUM	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
STRETCH	EX-7-STRETCH	VERIFIED	1	PASSED	FAILED (Good)	4/5	80.0%

EX-3-EASY (EASY) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

EX-5-MEDIUM (MEDIUM) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

EX-7-STRETCH (STRETCH) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-1**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

3. Template Compliance QA

Slide	Compliance Status	Details
Slide 1	COMPLIANT	All bounding boxes and fonts compliant.
Slide 2	COMPLIANT	All bounding boxes and fonts compliant.
Slide 3	COMPLIANT	All bounding boxes and fonts compliant.
Slide 4	COMPLIANT	All bounding boxes and fonts compliant.
Slide 5	COMPLIANT	All bounding boxes and fonts compliant.
Slide 6	COMPLIANT	All bounding boxes and fonts compliant.
Slide 7	COMPLIANT	All bounding boxes and fonts compliant.
Slide 8	COMPLIANT	All bounding boxes and fonts compliant.
Slide 9	COMPLIANT	All bounding boxes and fonts compliant.
Slide 10	COMPLIANT	All bounding boxes and fonts compliant.